Analysis Report

1 Treatment Effect on Hiring Outcomes

1.1 Blinding Disrupts Pre-Existing Demographic Premiums and Penalties

- **Blinding disrupts pre-existing system of demographic premiums and penalties.**
  - **Winners: Hispanic Female in Resume-Screening (Phase 1) and White Male in Technical-Evaluation (Phase 2).** Blinding removes the baseline *penalty* these groups received under non-blind evaluations. Their evaluation scores rise by +8.7% and +8.6% of a SD respectively.
  - **Losers: Hispanic Male in Phase 1 and Hispanic Female in Phase 2.** Blinding removes the baseline *premium* these groups received under non-blind evaluations. Their evaluation scores fall by -11.5% and -6.9% of a SD respectively.
- **Caveat:** Hiring is often an *ordinal ranking* problem — one candidate's gain corresponds to another's lost opportunity — so the absence of statistical significance on a given row does not mean the corresponding group is unaffected by blinding.

Table 1: Effect of Blinding on Overall Evaluation Score by Race-Gender

	Panel A: Resume-Screening	Panel B: Technical-Evaluation
	HR	Engineer
White Male	0.037	0.086***
	(0.058)	(0.028)
White Female	0.013	0.023
	(0.051)	(0.036)
Asian Male	0.014	−0.018
	(0.059)	(0.028)
Asian Female	−0.023	−0.029
	(0.048)	(0.029)
Hispanic Male	−0.115**	−0.003
	(0.048)	(0.030)
Hispanic Female	0.087*	−0.069**
	(0.052)	(0.034)
Display Position FE	✓	✓
Resume ID FE	✓	✓
Resume - Coding Test Version FE	×	×
Respondent ID FE	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓
Observations	2466	2772
R^2	0.423	0.743
Adj. R^2	0.409	0.737

* p < 0.1, ** p < 0.05, *** p < 0.01

Standard errors clustered by respondent.

1.1.1 Heterogeneity: Rejecting Traditional In-Group Favoritism

- **Findings:** Our sample of software engineers demonstrates a preference for the *out-group* (i.e., a preference for *under-represented groups*).
  - When demographic information is blinded, the scores of *concordant* (race/gender) applicants evaluated by Engineers rise by +6.9% of a SD ($p < 0.05$), while the scores of *discordant* applicants fall by -0.9% of a SD ($p < 0.05$).
  - White Males make up the majority of the evaluator pool for engineers, and their behavior drives the aggregate results.
- **Takeaway:** The majority gatekeepers were *actively penalizing* candidates who looked like themselves and assigning a *premium to evaluation scores of minority candidates* when such demographic information is visible. Enforcing blind hiring stops the majority group, White Males, from systematically boosting the minority candidates. This explains why, in our experiment, White Male applicants are the primary beneficiaries of blind hiring: they are "rescued" from the baseline penalty imposed by their own demographic engineer peers.

Table 2: Effect of Blinding on Overall Evaluation Score by Evaluator-Candidate Concordance (Race / Gender / Race+Gender)

	Panel A: Resume-Screening			Panel B: Technical-Evaluation		
	Race	Gender	Race+Gender	Race	Gender	Race+Gender
Concordant Applicant	−0.020	−0.010	−0.076	0.045*	0.029*	0.069**
	(0.042)	(0.027)	(0.068)	(0.023)	(0.015)	(0.032)
Nonconcordant Applicant	0.007	0.009	0.011	−0.014*	−0.030*	−0.009**
	(0.015)	(0.025)	(0.010)	(0.007)	(0.016)	(0.004)
Display Position FE	✓	✓	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×	×	×
Respondent ID FE	×	×	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓	✓	✓
Observations	2466	2466	2466	2772	2772	2772
R^2	0.421	0.421	0.421	0.742	0.742	0.742
Adj. R^2	0.408	0.408	0.408	0.737	0.737	0.737

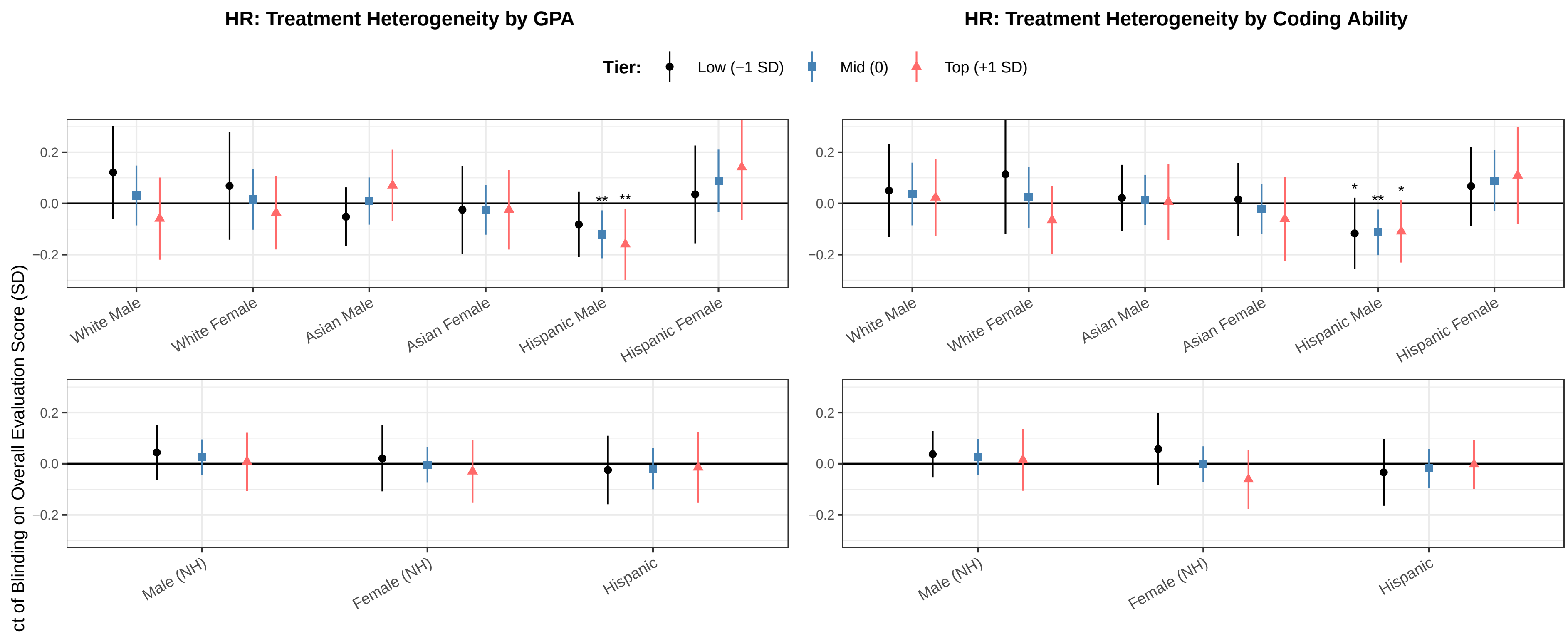
* p < 0.1, ** p < 0.05, *** p < 0.01

Standard errors clustered by respondent.

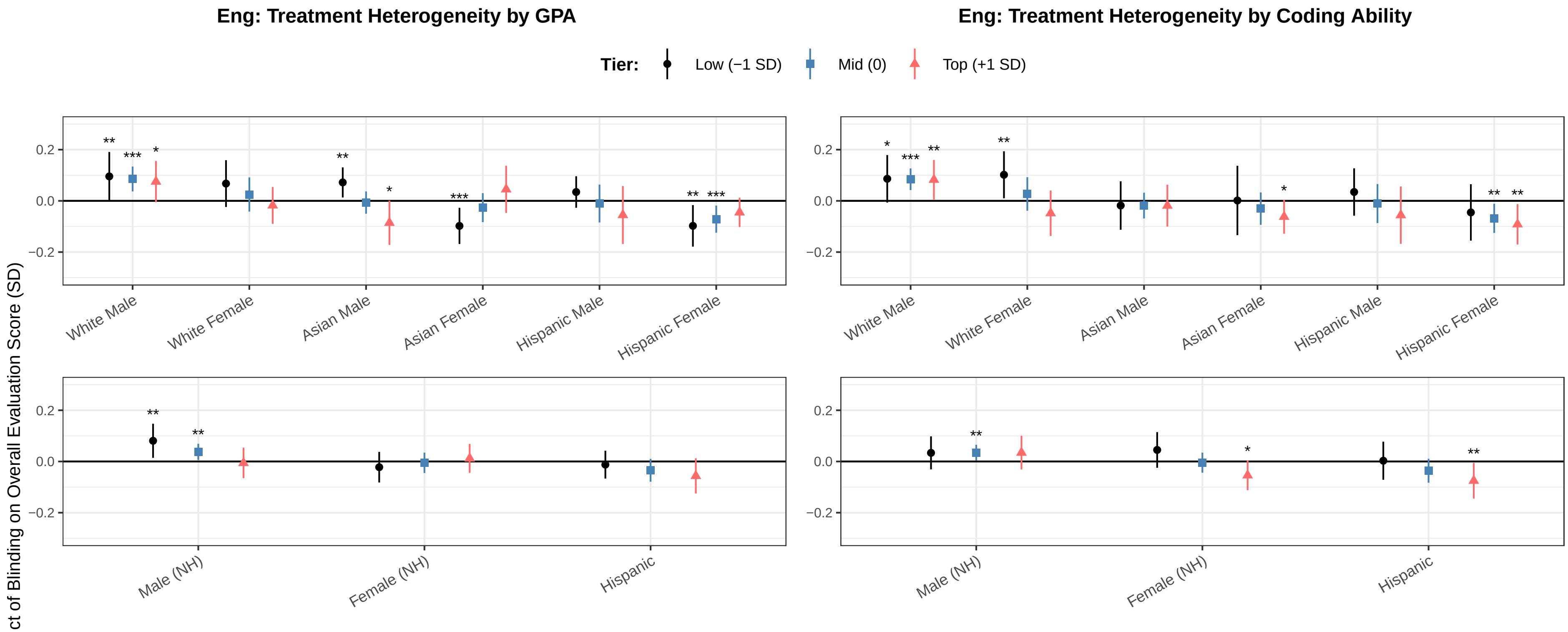
1.1.2 Heterogeneity: Where the Demographic Premium Applies (Effects by Candidate Quality)

- **Findings:** The blinding effect on evaluation scores diverges across visible candidate quality — GPA and Coding Ability (the success rate of the submitted code) — within the same demographic group. The implicit out-group premium documented in §1.1.1 is concentrated at the *top* of the ability distribution.
  - **Minorities.** In the *resume-screening* phase, blinding removes the **premium Hispanic Females** previously received at all low, mid, high ability tiers, as defined by visible ability cues (**GPA** and **code success rate**). In the *technical-evaluation* phase, blinding hurts **gender and racial minorities** by removing the **premium** they previously received under nonblind.
  - **Majority (White Male mirror).** Blinding raises **White Male** evaluation scores roughly *uniformly* across all ability tiers in both phases. The in-group discount evaluators apply to their own demographic under nonblind is a **flat demographic shift** — not conditional on the candidate's observable ability. Blinding removes that shift, helping **White Males** equally at low, mid, and high ability tiers.
  - **Takeaway.** Evaluators' demographic adjustments under nonblind are *threshold-based*: **minorities** receive a **premium only when** they clear a high visible-ability bar — concentrated at the *top* of the GPA / code-success-rate distribution. Evaluators appear to want to advance minority candidates **but not at the cost of quality**, reserving the premium for the strongest-on-paper minorities and giving nothing to the rest. The mirror **penalty** applied to **White Males** is structurally different: it is *unconditional*, a flat shift across ability tiers. Blinding removes both adjustments simultaneously.

HR Evaluators – Resume Screening Stage

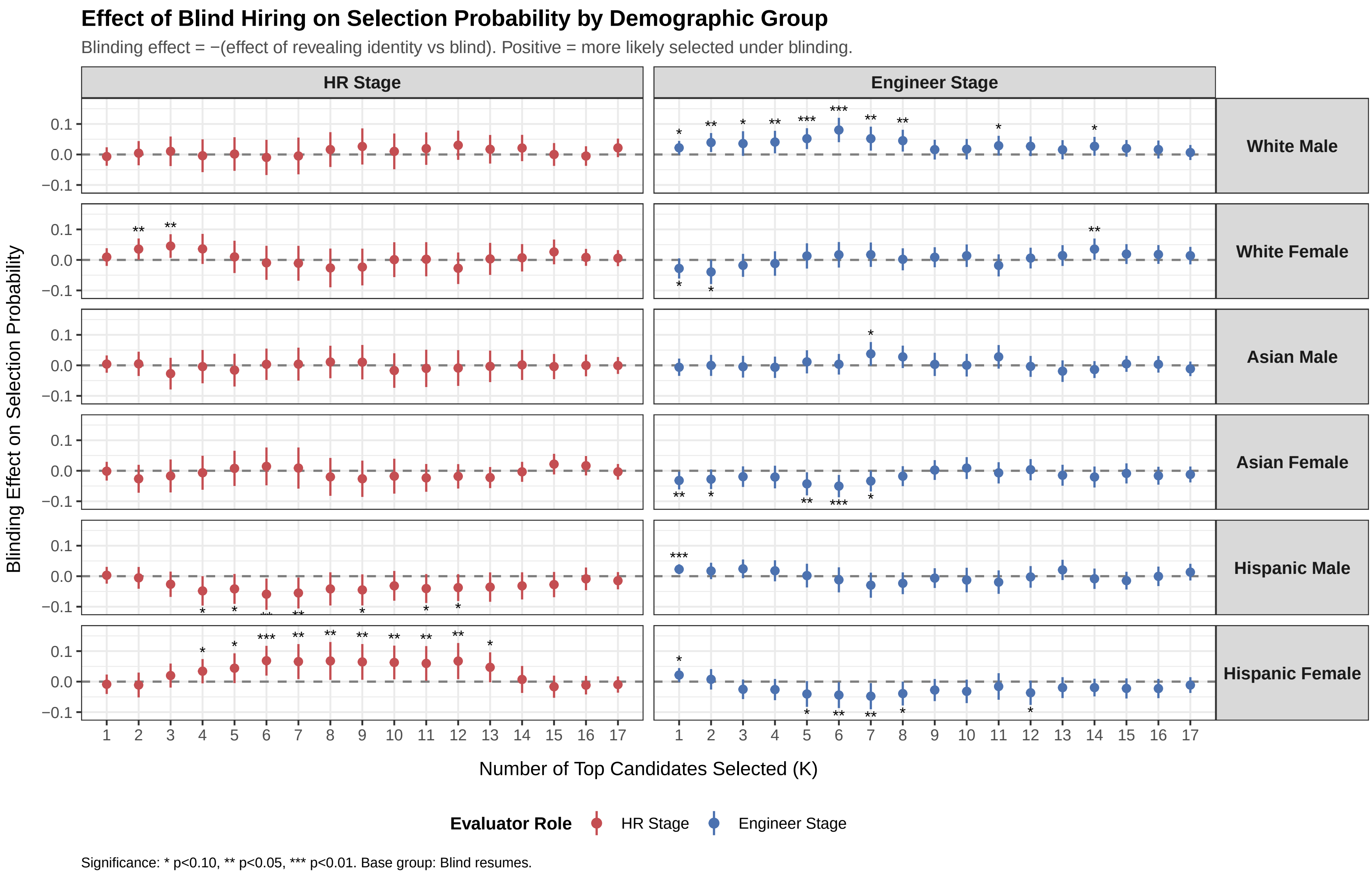


Engineer Evaluators – Technical Evaluation



1.2 Blinding Reshuffles Demographics of Hired (Composition Effect)

- **Findings:** Mirroring the per-resume rating effects in §1.1, blinding reshuffles who advances at each phase.
  - *Resume-screening phase:* **Hispanic Females** gain a higher pass rate to the next round; **Hispanic Males** lose.
  - *Technical-evaluation phase:* **White Males** gain at the hiring stage; **Asian Females** and **Hispanic Females** lose.



2 Allocation of Attention and Material Engagement

2.1 Determinants of Attention (Descriptive)

- **Findings:** Evaluators do **not** spend more time on stronger candidates — GPA, awards, top-firm experience, test-case performance, and resume–code mismatch are all statistically insignificant and economically negligible ($< 3\%$ of mean time per SD). The one meaningful determinant is **display-order fatigue**: each later position receives less time (HR -0.5s per position, $\sim$17% cumulative decline; Engineers -1.4s, $\sim$34%; both $p < 0.01$). The table uses display positions 4–18, dropping the first three as *warm-up*: the first resume alone averages 78s for HR and 171s for Engineers (vs. a $\sim$40s/$\sim$55s plateau) and engineer time stays elevated through position three.

Table 3: Determinants of Time on Page (Seconds)

	Panel A: Resume-Screening	Panel B: Technical-Evaluation
	(1)	(2)
GPA (Z)	−0.83 (0.60)	−0.53 (1.20)
# Awards (Z)	0.01 (0.68)	0.47 (1.03)
Top Experience	−0.89 (1.41)	3.88* (2.25)
Test Cases Passed (Z)		−0.14 (1.65)
Quality Mismatch I (Strong Resume, Weak Coding)		−0.69 (4.07)
Quality Mismatch II (Strong Coding, Weak Resume)		3.45 (2.64)
Display Position	−0.50*** (0.16)	−1.38*** (0.23)
Mean Time (Seconds)	40.1	57.1
Respondent ID FE	✓	✓
Observations	2050	2305
R^2	0.286	0.464

Within-evaluator (Respondent ID FE); SE clustered by respondent. Estimates in seconds, est* (SE) inline; * p<.10, ** p<.05, *** p<.01. Sample is display positions 4–18; the first three positions are dropped as warm-up. Blank cells mark signals HR never saw (test cases, code). Quality Mismatch I/II are above-/below-average splits on the paper composite (GPA, awards, firm prestige) versus test cases (Eng-only). URM-race and URM-gender (nonblind) are also insignificant (not shown).

- **Findings:** Material engagement responds to candidate quality — and it shows up in **opening the code, not the resume.** Engineers open the actual code more for stronger candidates: higher GPA (+1.5 pp per SD), higher test cases (+1.7 pp), and especially candidates who **code better than their resume credentials** (Quality Mismatch II — Strong Coding, Weak Resume: +3.2 pp, $\sim$+5%) — the "hidden gem" verification. Resume-opening responds to none of these, and time (above) responds to nothing. The reverse conflict (Quality Mismatch I — Strong Resume, Weak Coding), the candidates' assigned race/gender (Panel B, nonblind), and demographic×quality interactions are all null. Entered jointly, collinearity across the 18 candidates renders the individual signals insignificant, so each coefficient below is a focused regression.

Table 4: Determinants of Opening Rates (Engineer Phase Only)

	Open Resume	Open Code
Panel A: Quality, Mismatch & Fatigue (all rounds)		
GPA (Z)	0.7 (0.7)	1.5** (0.7)
Test Cases Passed (Z)	0.9 (0.6)	1.7** (0.8)
# Awards (Z)	-0.4 (0.6)	0.8 (0.6)
Top Experience	0.4 (1.2)	0.4 (1.1)
Quality Mismatch I (Strong Resume, Weak Coding)	-0.2 (1.6)	1.9 (1.7)
Quality Mismatch II (Strong Coding, Weak Resume)	1.0 (1.2)	3.2** (1.5)
Mean Open Rate (%)	64.5	63.9
Display Position Control	✓	✓
Respondent FE	✓	✓
Observations	2,772	2,772
Panel B: Demographics (Nonblind only)		
URM — Race	2.9 (2.1)	0.6 (1.9)
URM — Gender	1.6 (1.8)	2.3 (1.7)
Observations	1,332	1,332

Engineer phase only; outcome is opening a file (0/1). Each coefficient is a separate regression of the outcome on that signal (the two mismatch flags entered jointly; URM jointly), with Respondent ID FE and a display-position control; SE clustered by respondent. Coefficients in percentage points, est* (SE); * p<.10, ** p<.05, *** p<.01. Entered all together, collinearity across the 18 candidates renders these individually insignificant. Quality Mismatch I/II are above-/below-average splits on the paper-credential composite (GPA, awards, firm prestige) versus test cases (Strong Resume/Weak Coding and Strong Coding/Weak Resume; 3 and 4 of 18 candidates; reference = concordant). Demographic*times*quality and other interactions tested and null.

2.2 Treatment Effects on Engagement (Causal)

- **Findings:** Blinding does not change the **time** evaluators spend per resume in either phase. It does, however, change **what they engage with**: **Engineer evaluators** open candidates' resume PDF and coding scripts at a ∼10 pp higher rate under blinding than under nonblind.

Table 5: Effect of Blinding on Time Spent Per Resume

	Panel A: Resume-Screening		Panel B: Technical-Evaluation	
	(1)	(2)	(3)	(4)
Nonblind Group Mean (Seconds)	42.6	42.6	65.5	65.5
Blind Treatment	0.590	0.590	3.135	3.135
	(3.264)	(3.264)	(5.840)	(5.841)
Blind × Test Cases Passed (Z)		0.732		−0.076
		(1.243)		(1.627)
Display Position FE	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×
Respondent ID FE	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓
Observations	2460	2460	2765	2765
R^2	0.125	0.125	0.250	0.250
Adj. R^2	0.106	0.105	0.236	0.235

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Outcome in seconds. Standard errors clustered by respondent.

Table 6: Material Engagement (Engineer Only)

	Panel A: Open Resume		Panel B: Open Codes (1/2)	
	(1)	(2)	(3)	(4)
Nonblind Group Mean (Open Rate)	0.617	0.617	0.584	0.584
Blind Treatment	0.052	0.052	0.106*	0.106*
	(0.059)	(0.059)	(0.058)	(0.058)
Blind × Test Cases Passed (Z)		0.008		−0.019
		(0.011)		(0.015)
Display Position FE	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×
Respondent ID FE	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓
Observations	2772	2772	2772	2772
R^2	0.095	0.095	0.119	0.119
Adj. R^2	0.077	0.077	0.102	0.102

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Outcome is 0/1 (file opened). Standard errors clustered by respondent.

2.3 Signal Substitution – Re-weighting of Visible Resume Cues

Question: What visible resume signals does the evaluator lean on more (or less) under blinding?

Specification:

$$\text{Overall\_Evaluation\_Score (Z)}_{ij} \sim \sum_{s \in \mathcal{S}_{\text{role}}} \boldsymbol{\beta_s}\, \text{Signal}_s^i \times \mathbb{1}[\text{Blind}]_j \;\Big|\; \text{Resume\_ID FE}_i + \text{Respondent\_ID FE}_j + \text{Display\_Order FE}_{ij}$$

Two-way clustered SEs on (responseid, resume_id).

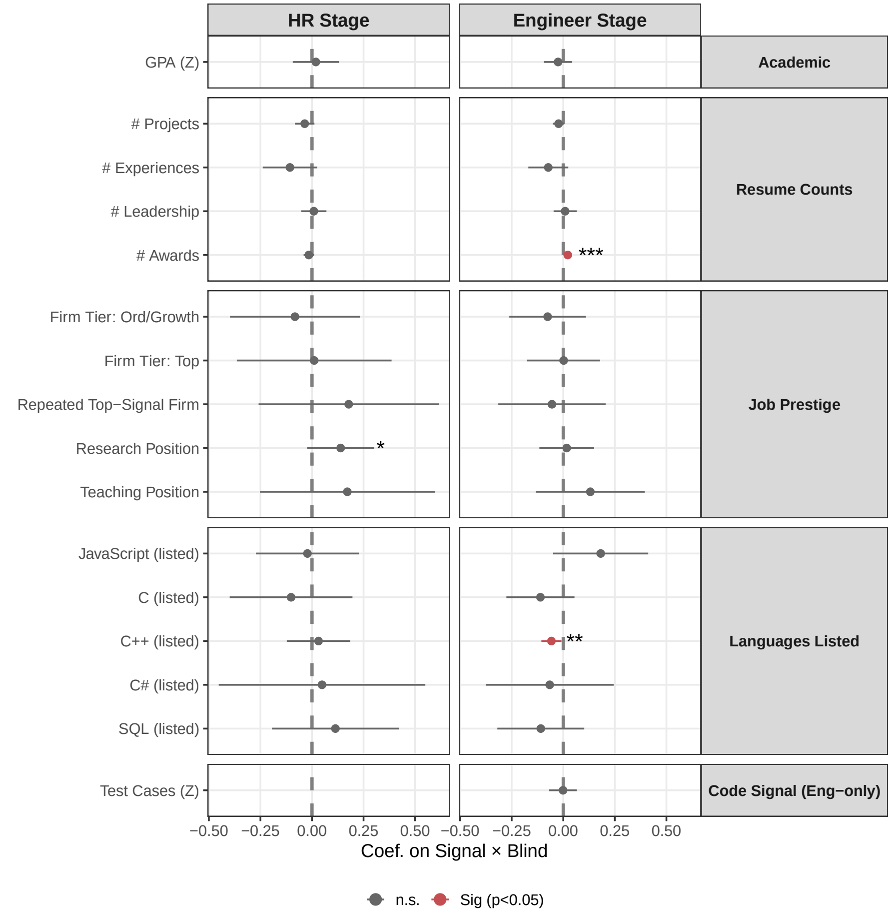
Table 7: Signal Substitution

Blinding: Change in Signal Weights on Evaluator-Assigned Overall Scores

	HR	Engineer
Academic		
GPA (Z)	0.019 (0.057)	−0.025 (0.035)
Resume Counts		
# Projects	−0.035 (0.024)	−0.023 (0.014)
# Experiences	−0.107 (0.067)	−0.073 (0.049)
# Leadership	0.009 (0.031)	0.009 (0.029)
# Awards	−0.015 (0.013)	0.022*** (0.008)
Job Prestige		
Firm Tier: Ord/Growth	−0.083 (0.161)	−0.076 (0.095)
Firm Tier: Top	0.011 (0.191)	0.002 (0.090)
Repeated Top-Signal Firm	0.178 (0.223)	−0.055 (0.133)
Research Position	0.139* (0.083)	0.017 (0.068)
Teaching Position	0.171 (0.216)	0.131 (0.135)
Languages Listed		
JavaScript (listed)	−0.022 (0.128)	0.181 (0.118)
C (listed)	−0.101 (0.152)	−0.111 (0.084)
C++ (listed)	0.032 (0.079)	−0.057** (0.025)
C# (listed)	0.049 (0.255)	−0.066 (0.158)
SQL (listed)	0.114 (0.157)	−0.109 (0.108)
Code Signal (Eng-only)		
Test Cases (Z)		−0.001 (0.034)
Display Position FE	✓	✓
Resume ID FE	✓	✓
Resume - Coding Test Version FE	×	×
Respondent ID FE	✓	✓
Respondent Age (Cont) and Demographic FEs	×	×
Obs.	2.466	2.772
R^2	0.424	0.743
Adj. R^2	0.377	0.722

Figure 1: Signal Substitution

Blinding: Change in Signal Weights on Evaluator-Assigned Overall Scores



2.4 Re-weighing of Internal Rubric – Coding / Fit / Stay Dimension Aggregation

Question: Do the weights evaluators place on the three sub-dimension ratings (Coding Skill, Fit, Stay) shift under blinding?

Specification:

$$\text{Overall\_Evaluation\_Score (Z)}_{ij} \sim \sum_{d \in \{\text{Coding\_Skill, Fit, Stay}\}} \left(\beta_d + \boldsymbol{\gamma_d}\, \mathbb{1}[\text{Blind}]_j\right) \text{Score}_d^i$$

$$\Big|\ \text{Resume\_ID FE}_i + \text{Respondent\_ID FE}_j + \text{Display\_Order FE}_{ij}$$

Two-way clustered SEs on (responseid, resume_id).

Table 8: Internal Rubric Shift

	HR	Engineer
Sub-dimension level (nonblind baseline)		
Coding (Z)	0.490*** (0.046)	0.537*** (0.038)
Fit (Z)	0.267*** (0.047)	0.222*** (0.039)
Stay (Z)	0.139*** (0.031)	0.077*** (0.018)
Re-weighting under Blind Evaluation		
Coding × Blind	0.027 (0.059)	−0.046 (0.056)
Fit × Blind	0.007 (0.055)	0.037 (0.057)
Stay × Blind	−0.064 (0.039)	0.003 (0.028)
Display Position FE	✓	✓
Resume ID FE	✓	✓
Resume - Coding Test Version FE	×	×
Respondent ID FE	✓	✓
Respondent Age (Cont) and Demographic FEs	×	×
Obs.	2.376	2.736
R^2	0.788	0.889
Adj. R^2	0.771	0.880

3 Signal Validity: Are Visible Signals Predictive of True Coding Ability?

(NEED TO RE-RUN WITH ALL 69 STUDENT RESUMES)

Why this matters. §2.2 shows that under blinding, evaluators reweight their attention onto visible resume signals — but a positive weight-shift only counts as rational signal substitution if those signals *actually predict* true productivity. If the signals blinding pushes evaluators toward have zero validity, the §2.2 effect is just *credentialism* (the flight-to-credentials hypothesis). This section provides the validity benchmark.

Question: Among the visible resume signals evaluators rely on, which actually predict true productivity?

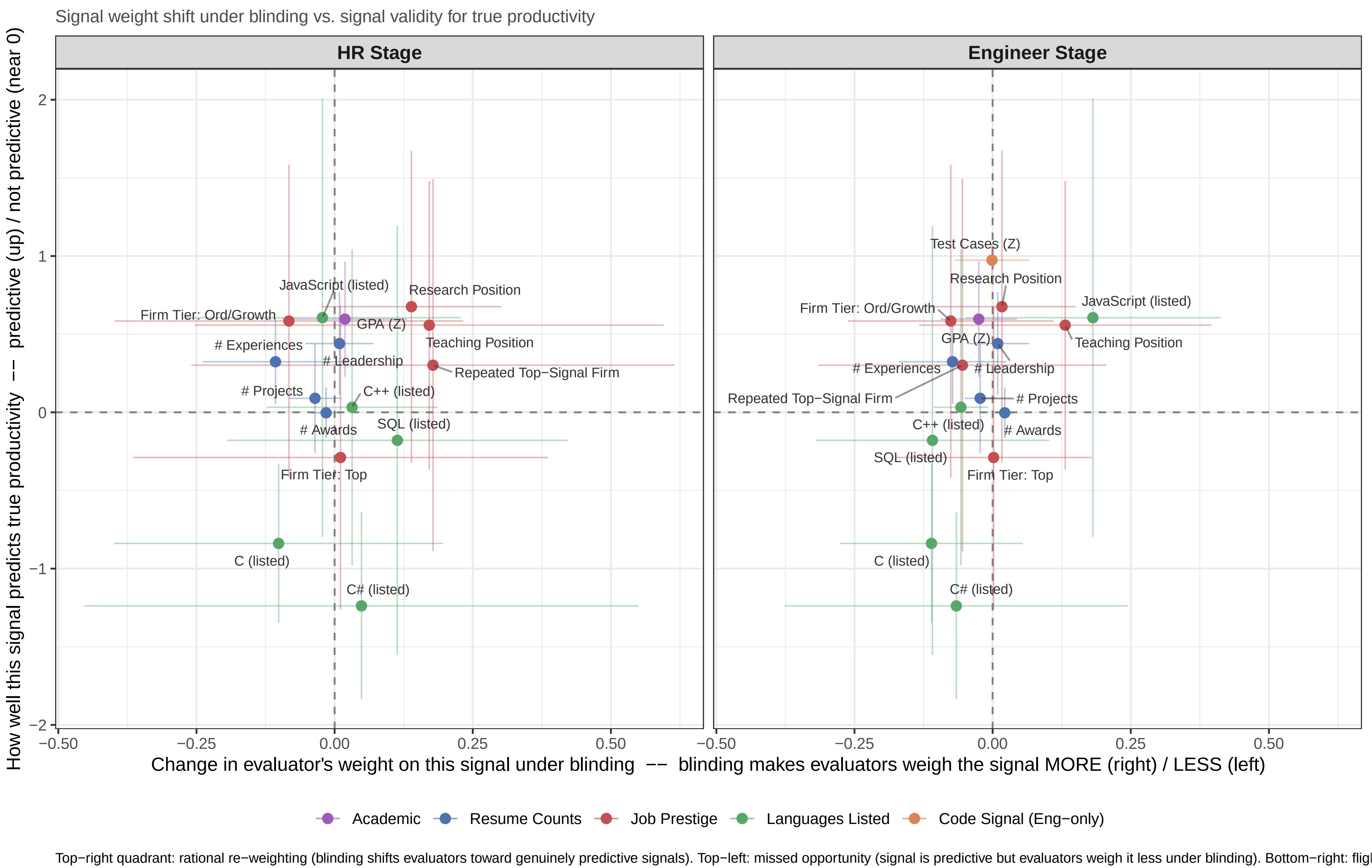
Specification:

$$\text{True\_Productivity\_Score (Z)}_i \sim \alpha + \boldsymbol{\gamma_s}\,\text{Signal}^i_s + \varepsilon_i, \quad i = 1, \ldots, 18$$

One univariate OLS per signal at the resume level; HC1 robust SE.

Table 9: Signal Validity — univariate cross-resume slope of True Productivity Score (Z) on each visible signal.

Signal	Estimate	R^2	Obs.
Academic			
GPA (Z)	0.596*** (0.187)	0.336	18
Resume Counts			
# Projects	0.090 (0.177)	0.016	18
# Experiences	0.324** (0.137)	0.200	18
# Leadership	0.440** (0.167)	0.107	18
# Awards	-0.003 (0.082)	0.000	18
Job Prestige			
Firm Tier: Ord/Growth	0.584 (0.510)	0.068	18
Firm Tier: Top	-0.289 (0.495)	0.021	18
Repeated Top-Signal Firm	0.302 (0.607)	0.016	18
Research Position	0.676 (0.508)	0.109	18
Teaching Position	0.558 (0.470)	0.077	18
Languages Listed			
JavaScript (listed)	0.605 (0.715)	0.051	18
C (listed)	-0.839*** (0.260)	0.037	18
C++ (listed)	0.032 (0.515)	0.000	18
C# (listed)	-1.239*** (0.304)	0.213	18
SQL (listed)	-0.179 (0.700)	0.003	18
Code Signal (Eng-only)			
Test Cases (Z)	0.974*** (0.053)	0.948	18



Signal weight shift under blinding vs. signal validity for true productivity
HR Stage
Engineer Stage
How well this signal predicts true productivity –– predictive (up) / not predictive (near 0)
Change in evaluator's weight on this signal under blinding –– blinding makes evaluators weigh the signal MORE (right) / LESS (left)
2
1
0
−1
−2
−0.50
−0.25
0.00
0.25
0.50
JavaScript (listed)
Research Position
Firm Tier: Ord/Growth
GPA (Z)
Experiences
Teaching Position
Leadership
Repeated Top−Signal Firm
Projects
C++ (listed)
Awards
SQL (listed)
Firm Tier: Top
C (listed)
C# (listed)
Test Cases (Z)
Academic
Resume Counts
Job Prestige
Languages Listed
Code Signal (Eng−only)
Top−right quadrant: rational re−weighting (blinding shifts evaluators toward genuinely predictive signals). Top−left: missed opportunity (signal is predictive but evaluators weigh it less under blinding). Bottom−right: fligl

4 Individual Response to Institutional Policy (Institutional Crowding Out or Compensatory Bias)

Takeaway / Summary.

- **The Puzzle of Individual Ideology:** The two-way relationship between an evaluator's personal DEI priority and their evaluation scores is surprisingly *muted and statistically noisy*. At first glance, personal ideology appears to have little impact on how blind hiring reshuffles outcomes.
  - **Compensatory Bias** (Firms with NO Diversity Mandate, see red lines): When a firm lacks an explicit objective policy targeting URMs, highly pro-DEI HR professionals engage in *compensatory bias* — filling the perceived institutional void by informally inflating minority candidates' scores in the baseline.
  - **Institutional Crowding-Out** (Firms WITH a Diversity Mandate, see blue lines): Conversely, when the firm explicitly mandates URM targeting, this formal policy *crowds out* the evaluator's informal efforts; pro-DEI HR professionals *mute* their premium to under-represented minorities' scores.
- (Note the *divergence between HR and Engineer results*, particularly in the triple-interaction columns.)

Note on Key Index Creation.

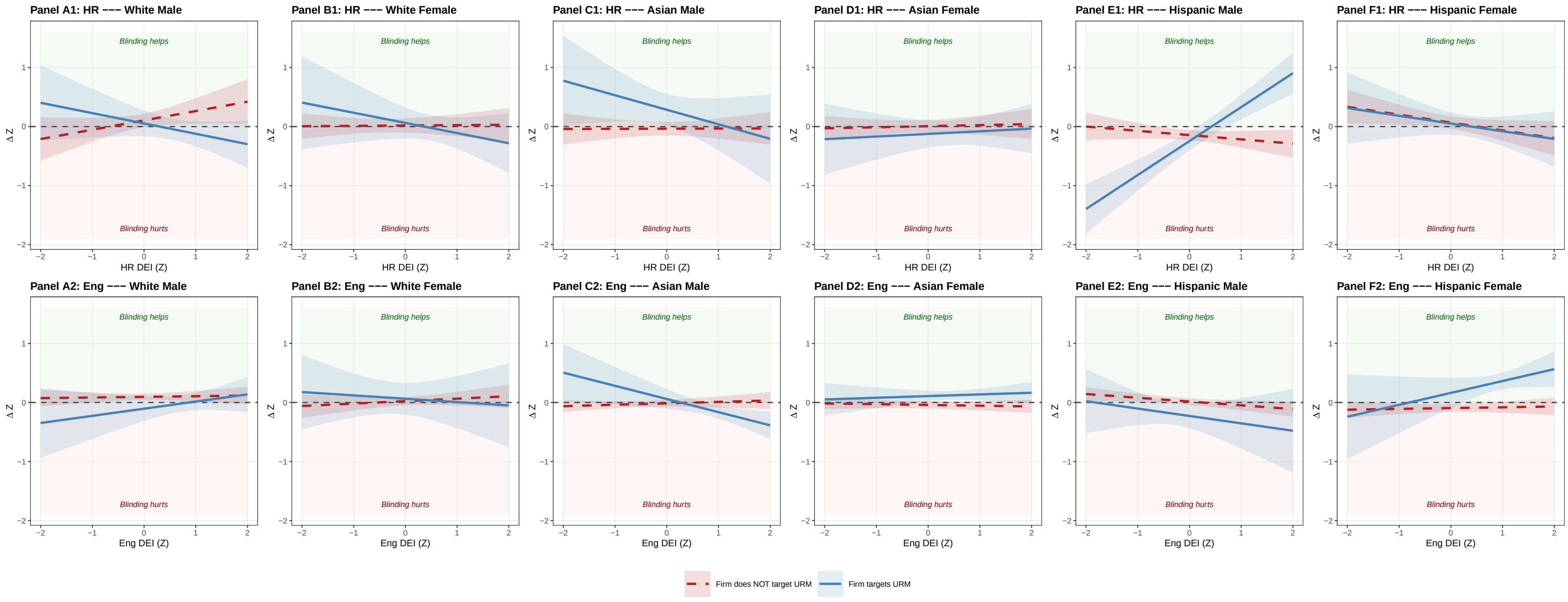
- *Individual DEI Index (Z)*: Evaluator's personal DEI orientation, condensed into a single z-scored score. (Low = Anti-DEI, High = Pro-DEI)
  - Mathematical expression: Index $= z\big(\frac{1}{5}\sum_{j=1}^{5} z(x_j)\big)$, z-scored within (`role`, `treat`) cells.
  - `pro_dei_race` — stated agreement that race DEI is important in hiring (Likert).
  - `pro_dei_gender` — stated agreement that gender DEI is important in hiring (Likert).
  - `pro_diverse` — stated importance of working at a company with a diverse workforce (Likert).
  - `pro_hiring_dei` — stated belief that DEI initiatives help firms hire better candidates (Likert).
  - `div_priority` — revealed-priority rank of "Diversity and Representation" in the seven-goals question.
- *Firm Targets Fem/His (0/1)*: Indicator for firm policy — whether the evaluator's firm formally targets Female or Hispanic candidates in hiring.
  - Mathematical expression: Idx $= \mathbb{1}\{\texttt{firm\_targets\_female} = 1 \lor \texttt{firm\_targets\_hispanic} = 1\}$.
  - `firm_targets_female` — 1 if option 6 ("Female") was selected on `s5_policy_diversity` (multi-select).
  - `firm_targets_hispanic` — 1 if option 3 ("Hispanic or Latino") was selected on the same multi-select.
- *Firm DEI Attn Index (Z)*: Respondent's perception of how much attention their firm currently pays to race and gender DEI in hiring, condensed into a single z-scored score.
  - Mathematical expression: Index $= z\big(\frac{1}{2}\big(z(\texttt{race\_attn}) + z(\texttt{gender\_attn})\big)\big)$, z-scored within (`role`, `treat`) cells.
  - `s5_view_dei_race_f` — Likert: respondent's perception of firm's current attention to race DEI in hiring.
  - `s5_view_dei_gender_f` — Likert: respondent's perception of firm's current attention to gender DEI in hiring.

4.1 By Six Demographics

Table 10: DEI Heterogeneity in the Effect of Blinding on Overall Evaluation Score, by Race-Gender (6 cells): Simple, Triple-with-Firm-Targets-URM, Triple-with-Firm-Pays-DEI-Attention.

	Simple DEI Heterogeneity		Triple: Firm Targets URM		Triple: Firm Pays DEI Attention	
	HR	Engineer	HR	Engineer	HR	Engineer
White Male	0.059 (0.053)	0.085*** (0.028)	0.105* (0.059)	0.095*** (0.029)	0.128** (0.062)	0.054* (0.030)
White Female	0.011 (0.052)	0.022 (0.035)	0.018 (0.061)	0.023 (0.036)	−0.001 (0.067)	0.032 (0.036)
Asian Male	0.015 (0.060)	−0.017 (0.028)	−0.036 (0.058)	−0.014 (0.030)	0.072 (0.068)	−0.028 (0.034)
Asian Female	−0.021 (0.048)	−0.030 (0.029)	0.006 (0.055)	−0.041 (0.031)	−0.027 (0.060)	−0.020 (0.042)
Hispanic Male	−0.109** (0.046)	−0.003 (0.029)	−0.144*** (0.043)	0.016 (0.029)	−0.179*** (0.053)	0.030 (0.028)
Hispanic Female	0.064 (0.047)	−0.070** (0.033)	0.068 (0.055)	−0.095*** (0.033)	0.032 (0.055)	−0.081* (0.047)
White Male × Individual DEI Index (Z)	0.098 (0.079)	0.010 (0.032)	0.158* (0.091)	0.011 (0.034)	0.166** (0.066)	0.040 (0.045)
White Female × Individual DEI Index (Z)	−0.017 (0.051)	0.037 (0.047)	0.006 (0.056)	0.041 (0.048)	0.000 (0.069)	−0.011 (0.046)
Asian Male × Individual DEI Index (Z)	−0.005 (0.064)	0.013 (0.026)	0.002 (0.062)	0.024 (0.028)	−0.021 (0.066)	0.044 (0.048)
Asian Female × Individual DEI Index (Z)	0.009 (0.046)	−0.006 (0.022)	0.018 (0.052)	−0.013 (0.022)	−0.006 (0.059)	0.008 (0.050)
Hispanic Male × Individual DEI Index (Z)	0.020 (0.062)	−0.076*** (0.029)	−0.072 (0.057)	−0.064** (0.028)	−0.033 (0.069)	−0.096** (0.041)
Hispanic Female × Individual DEI Index (Z)	−0.135** (0.062)	0.037 (0.034)	−0.133* (0.069)	0.013 (0.033)	−0.152*** (0.056)	0.012 (0.065)
White Male × Firm Targets Fem/His (0/1)			−0.052 (0.128)	−0.201* (0.111)		
White Female × Firm Targets Fem/His (0/1)			0.044 (0.149)	0.040 (0.142)		
Asian Male × Firm Targets Fem/His (0/1)			0.321** (0.148)	0.073 (0.090)		
Asian Female × Firm Targets Fem/His (0/1)			−0.131 (0.131)	0.150*** (0.055)		
Hispanic Male × Firm Targets Fem/His (0/1)			−0.101 (0.093)	−0.244** (0.109)		
Hispanic Female × Firm Targets Fem/His (0/1)			−0.017 (0.112)	0.257* (0.135)		
White Male × Firm DEI Attn Index (Z)					−0.036 (0.108)	0.011 (0.027)
White Female × Firm DEI Attn Index (Z)					−0.111 (0.068)	0.032 (0.043)
Asian Male × Firm DEI Attn Index (Z)					0.129* (0.074)	−0.046 (0.041)
Asian Female × Firm DEI Attn Index (Z)					−0.017 (0.076)	−0.042 (0.037)
Hispanic Male × Firm DEI Attn Index (Z)					0.166** (0.069)	0.021 (0.026)
Hispanic Female × Firm DEI Attn Index (Z)					−0.116 (0.080)	0.033 (0.052)
White Male × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.333** (0.152)	0.110 (0.111)		
White Female × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.178 (0.164)	−0.098 (0.163)		
Asian Male × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.249 (0.192)	−0.247*** (0.091)		
Asian Female × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.026 (0.130)	0.041 (0.059)		
Hispanic Male × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.648*** (0.106)	−0.061 (0.154)		
Hispanic Female × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.002 (0.147)	0.188 (0.128)		
White Male × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					−0.222* (0.125)	0.024 (0.030)
White Female × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.014 (0.090)	0.006 (0.041)
Asian Male × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					−0.219** (0.089)	0.003 (0.027)
Asian Female × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.037 (0.078)	−0.022 (0.029)
Hispanic Male × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.159* (0.085)	−0.046** (0.022)
Hispanic Female × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.219** (0.104)	0.041 (0.046)
Display Position FE	✓	✓	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×	×	×
Respondent ID FE	×	×	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2160	2358

Standard errors clustered by respondent.



Marginal-effects "scissors" plot built from the §2.1 columns 3-4 (HR & Engineer triple with Firm Targets Fem/His), per race-gender. Top row: HR evaluators (Panels A1–F1). Bottom row: Engineer evaluators (Panels A2–F2). Lines: predicted effect of blinding on each group's overall score as a function of the rater's Individual DEI Index (Z); shaded bands are 95% CIs (SE clustered by respondent).

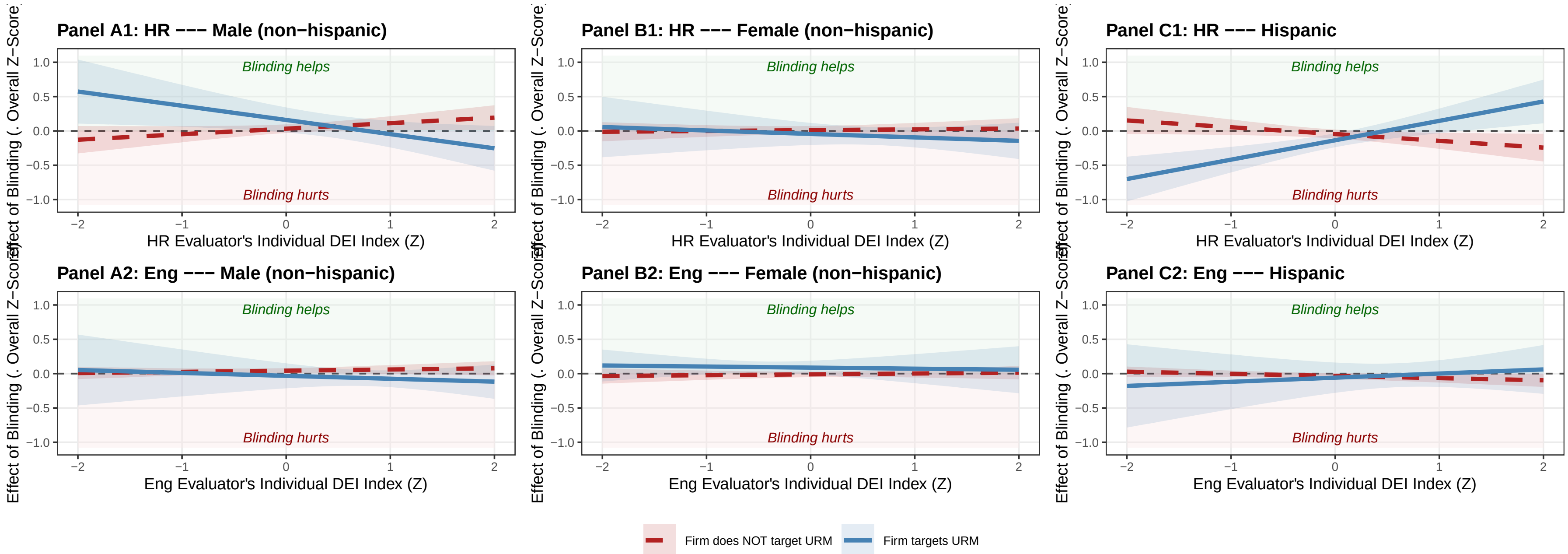
4.2 By Trichotomy (Majority vs Gender Minority vs Racial Minority)

Table 11: DEI Heterogeneity in the Effect of Blinding on Overall Evaluation Score, by URM Axis (Male NH / Female NH / Hispanic): Simple, Triple-with-Firm-Targets-URM, Triple-with-Firm-Pays-DEI-Attention.

	Simple DEI Heterogeneity		Triple: Firm Targets URM		Triple: Firm Pays DEI Attention	
	HR	Engineer	HR	Engineer	HR	Engineer
Male (non-hispanic)	0.036	0.036*	0.032	0.043**	0.099**	0.014
	(0.034)	(0.019)	(0.032)	(0.019)	(0.038)	(0.019)
Female (non-hispanic)	−0.007	−0.005	0.011	−0.011	−0.016	0.005
	(0.028)	(0.020)	(0.031)	(0.021)	(0.034)	(0.025)
Hispanic	−0.030	−0.033	−0.046	−0.034	−0.084***	−0.021
	(0.032)	(0.021)	(0.033)	(0.021)	(0.032)	(0.026)
Male (NH) × Individual DEI Index (Z)	0.048	0.012	0.080*	0.017	0.074**	0.042
	(0.043)	(0.022)	(0.046)	(0.023)	(0.036)	(0.032)
Female (NH) × Individual DEI Index (Z)	−0.002	0.014	0.012	0.012	−0.001	0.000
	(0.031)	(0.024)	(0.034)	(0.025)	(0.037)	(0.032)
Hispanic × Individual DEI Index (Z)	−0.048	−0.027	−0.099**	−0.031	−0.079*	−0.045
	(0.046)	(0.019)	(0.048)	(0.019)	(0.043)	(0.038)
Male (NH) × Firm Targets Fem/His (0/1)			0.127	−0.075		
			(0.099)	(0.095)		
Female (NH) × Firm Targets Fem/His (0/1)			−0.055	0.098*		
			(0.089)	(0.055)		
Hispanic × Firm Targets Fem/His (0/1)			−0.091	−0.025		
			(0.061)	(0.114)		
Male (NH) × Firm DEI Attn Index (Z)					0.045	−0.017
					(0.055)	(0.023)
Female (NH) × Firm DEI Attn Index (Z)					−0.064	−0.006
					(0.039)	(0.025)
Hispanic × Firm DEI Attn Index (Z)					0.020	0.025
					(0.052)	(0.026)
Male (NH) × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.287***	−0.060		
			(0.102)	(0.094)		
Female (NH) × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			−0.062	−0.028		
			(0.089)	(0.074)		
Hispanic × Individual DEI Index (Z) × Firm Targets Fem/His (0/1)			0.382***	0.091		
			(0.093)	(0.115)		
Male (NH) × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					−0.221***	0.013
					(0.069)	(0.022)
Female (NH) × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.027	−0.009
					(0.055)	(0.022)
Hispanic × Individual DEI Index (Z) × Firm DEI Attn Index (Z)					0.195***	−0.005
					(0.071)	(0.023)
Display Position FE	✓	✓	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×	×	×
Respondent ID FE	×	×	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓	✓	✓
Observations	2466	2772	2466	2772	2160	2358

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Standard errors clustered by respondent.



Marginal-effects "scissors" plot built from the §2.2 columns 3-4 (HR & Engineer triple with Firm Targets Fem/His). Top row: HR evaluators (Panels A1, B1, C1). Bottom row: Engineer evaluators (Panels A2, B2, C2). Lines: predicted effect of blinding on each group's overall score as a function of the rater's Individual DEI Index (Z); shaded bands are 95% CIs from the linear combination of axis-triple coefficients (SE clustered by respondent).

5 Inter-Departmental Backlash and Compensation: Signed Directional Index

Spec. Triple-interaction model with a **signed, DEI-specific, directional** perception index per role: positive end captures the role-appropriate "misalignment in the bad direction'' (HR overreach for Eng; Eng undercare for HR); negative end captures the opposite direction. Both directions are continuous, not thresholded.

Survey directional structure (used directly). The multi-select options `aligndiff{role}_1..6` record which DEI dimensions the respondent flagged, and the items themselves are directional:

- Options 1 / 4: counterpart places *higher* priority on gender / racial diversity (rebellion-relevant for Eng; over-care-relevant for HR).
- Options 2 / 5: counterpart places *lower* priority on gender / racial diversity (under-care-relevant for HR; opposite-direction signal for Eng).
- Options 3 / 6: counterpart places priority on *different* groups (no direction — ignored here).

Index construction (per role; respondent-level then merged back):

- **Eng (HR Overreach direction).** Likert: `s5_hr_influence` (1–4, "to what extent does HR push DEI in hiring"; higher = more push). Directional count: $(\texttt{aligndiffeng}_1 + \texttt{aligndiffeng}_4) - (\texttt{aligndiffeng}_2 + \texttt{aligndiffeng}_5)$ (positive = HR cares more on race/gender DEI; negative = HR cares less). Composite: `eng_dir = s5_hr_influence + 0.5` $\cdot$ `directional_count`, then z-scored within Eng.
- **HR (Eng Undercare direction).** Likert: `-s5_engineer_care` (negated 0–4 "to what extent does Eng prioritize DEI"; higher negated = Eng cares less). Directional count: $(\texttt{aligndiffhr}_2 + \texttt{aligndiffhr}_5) - (\texttt{aligndiffhr}_1 + \texttt{aligndiffhr}_4)$ (positive = Eng cares less on race/gender DEI; negative = Eng cares more). Composite: `hr_dir = -s5_engineer_care + 0.5` $\cdot$ `directional_count`, then z-scored within HR.
- `misalign_z` = `eng_dir_z` for Eng rows / `hr_dir_z` for HR rows, used as `perc_role` in the regressions. Sample inclusion: drop only respondents NA on the Likert anchor (`s5_hr_influence` for Eng / `s5_engineer_care` for HR).

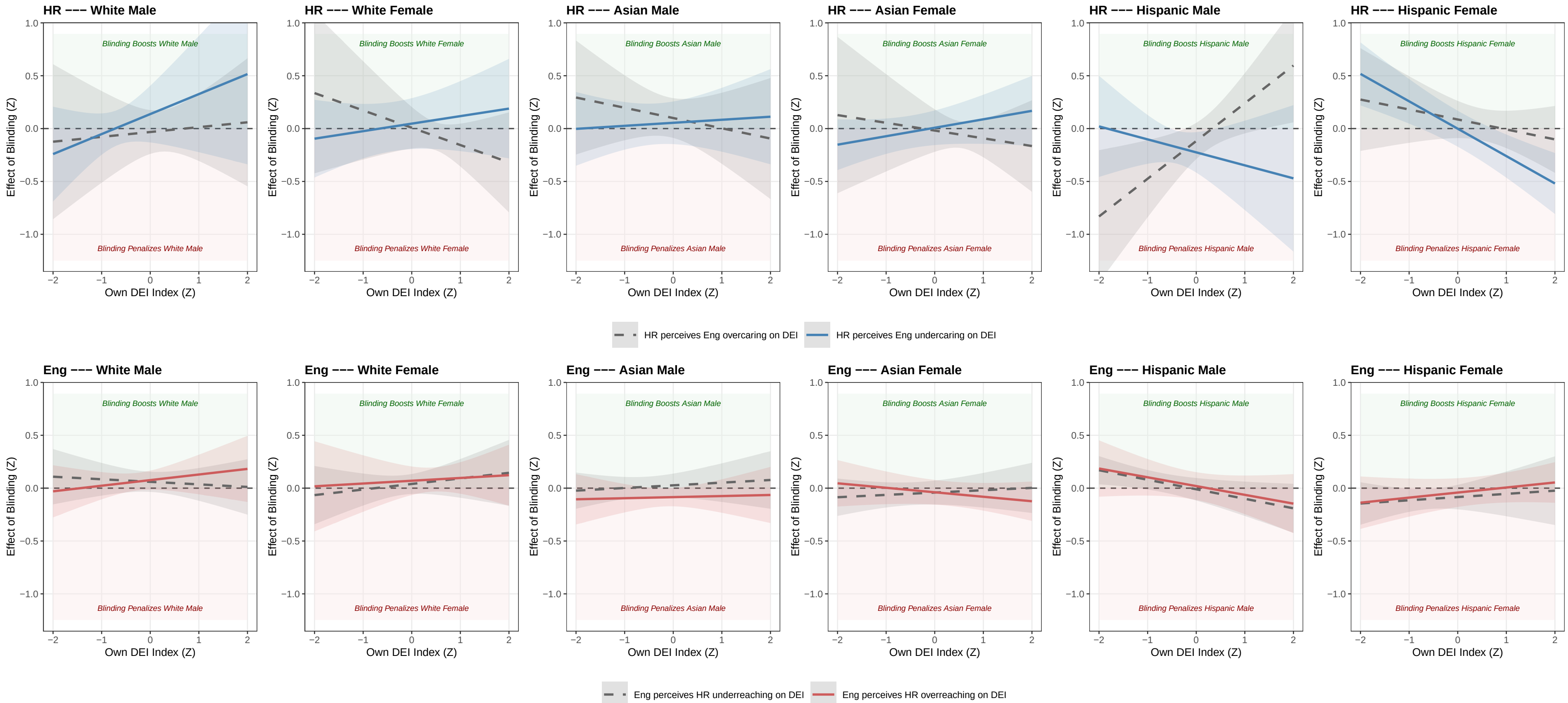
Substantive interpretation. `perc_role` > 0 = `rebellion conditions'' for Eng /`compensation conditions'' for HR; `perc_role` < 0 = the opposite direction (Eng who think HR underreaches; HR who think Eng overcares). Triple-interaction slopes can therefore be read as: *how does the effect of blinding on group X change as the rater moves from "counterpart cares too little on DEI" to "counterpart cares too much on DEI"?*

5.1 By Six Demographics

	Simple Dir Het.		Triple: Dir x Own DEI	
	HR	Engineer	HR	Engineer
White Male	0.016 (0.071)	0.086*** (0.027)	0.052 (0.095)	0.068** (0.030)
White Female	−0.020 (0.066)	0.050 (0.038)	0.027 (0.076)	0.055 (0.045)
Asian Male	0.051 (0.051)	−0.032 (0.035)	0.077 (0.053)	−0.029 (0.041)
Asian Female	−0.036 (0.051)	−0.055 (0.034)	−0.006 (0.066)	−0.040 (0.036)
Hispanic Male	−0.083 (0.060)	0.005 (0.050)	−0.171** (0.079)	0.005 (0.050)
Hispanic Female	0.081 (0.067)	−0.058 (0.042)	0.042 (0.069)	−0.063 (0.052)
White Male × Dir Idx (Z)	0.008 (0.055)	0.022 (0.049)	0.085 (0.077)	0.008 (0.036)
White Female × Dir Idx (Z)	−0.008 (0.064)	0.030 (0.033)	0.019 (0.082)	0.015 (0.039)
Asian Male × Dir Idx (Z)	−0.032 (0.081)	−0.050 (0.032)	−0.023 (0.084)	−0.056* (0.030)
Asian Female × Dir Idx (Z)	−0.019 (0.053)	−0.009 (0.033)	0.013 (0.066)	0.001 (0.046)
Hispanic Male × Dir Idx (Z)	−0.002 (0.051)	−0.022 (0.034)	−0.054 (0.049)	0.015 (0.035)
Hispanic Female × Dir Idx (Z)	0.060 (0.056)	0.038 (0.039)	−0.044 (0.055)	0.021 (0.038)
White Male × Own DEI Index (Z)			0.118 (0.121)	0.014 (0.061)
White Female × Own DEI Index (Z)			−0.046 (0.109)	0.039 (0.057)
Asian Male × Own DEI Index (Z)			−0.034 (0.082)	0.018 (0.047)
Asian Female × Own DEI Index (Z)			0.003 (0.083)	−0.010 (0.034)
Hispanic Male × Own DEI Index (Z)			0.117 (0.104)	−0.087** (0.039)
Hispanic Female × Own DEI Index (Z)			−0.177*** (0.059)	0.039 (0.041)
White Male × Dir Idx (Z) × Own DEI Index (Z)			0.072 (0.107)	0.039 (0.024)
White Female × Dir Idx (Z) × Own DEI Index (Z)			0.117* (0.061)	−0.013 (0.055)
Asian Male × Dir Idx (Z) × Own DEI Index (Z)			0.063 (0.078)	−0.007 (0.030)
Asian Female × Dir Idx (Z) × Own DEI Index (Z)			0.077 (0.074)	−0.032 (0.027)
Hispanic Male × Dir Idx (Z) × Own DEI Index (Z)			−0.240** (0.098)	0.004 (0.033)
Hispanic Female × Dir Idx (Z) × Own DEI Index (Z)			−0.082 (0.054)	0.009 (0.036)
Display Position FE	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×
Respondent ID FE	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓
Observations	2034	2124	2034	2124
R^2	0.433	0.739	0.441	0.740
Adj. R^2	0.414	0.730	0.418	0.730
R^2 Within	0.003	0.007	0.016	0.011
Adj. R^2 Within	−0.004	0.000	0.002	−0.002

Dir Idx (Z) = signed directional misalignment index: for Eng, HR Overreach direction (Likert + 0.5 times signed multi-select); for HR, Eng Undercare direction. Z-scored within role. Positive = rebellion conditions for Eng / compensation conditions for HR; negative = opposite. n = 113 HR, 118 Eng. Coefs flipped post-fit so positive = effect of blinding raises the score. Two-way cluster on responseid and resume index.

Dir Idx (signed) × Own DEI Index — Per Race-Gender (6 cells)



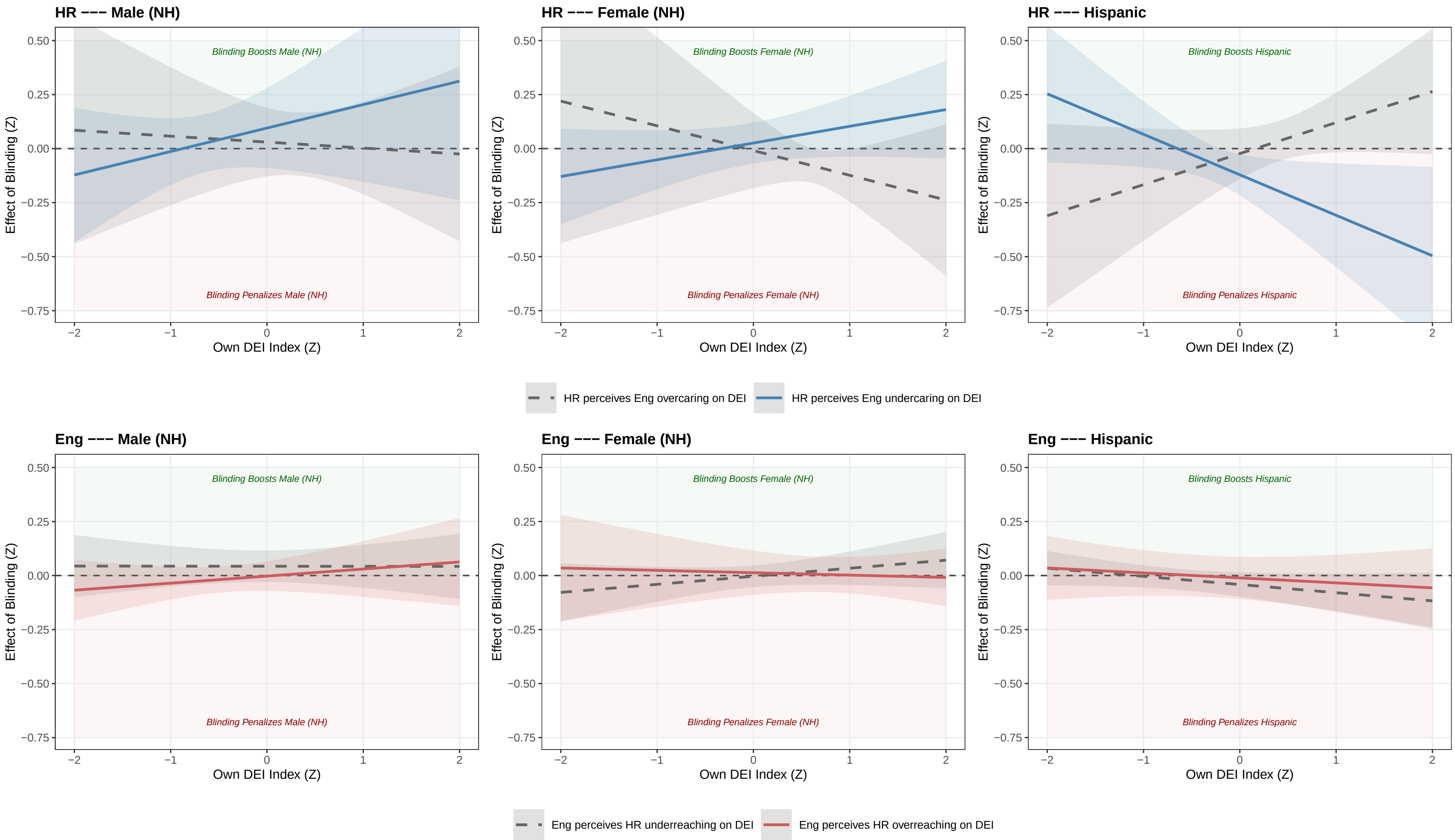
Top row: HR (high Dir = perceives Eng undercare = compensation conditions). Bottom row: Eng (high Dir = perceives HR overreach = rebellion conditions). Bands = 95% CI.

5.2 By Trichotomy (Majority vs Gender Minority vs Racial Minority)

	Simple Dir Het.		Triple: Dir x Own DEI	
	HR	Engineer	HR	Engineer
Male (NH)	0.033 (0.043)	0.028 (0.020)	0.063 (0.059)	0.020 (0.024)
Female (NH)	−0.029 (0.041)	−0.006 (0.025)	0.008 (0.055)	0.005 (0.033)
Hispanic	−0.005 (0.044)	−0.024 (0.030)	−0.072 (0.049)	−0.026 (0.033)
Male (NH) × Dir Idx (Z)	−0.011 (0.055)	−0.013 (0.034)	0.033 (0.065)	−0.023 (0.027)
Female (NH) × Dir Idx (Z)	−0.013 (0.036)	0.010 (0.017)	0.017 (0.046)	0.008 (0.025)
Hispanic × Dir Idx (Z)	0.025 (0.029)	0.004 (0.029)	−0.049** (0.023)	0.015 (0.024)
Male (NH) × Own DEI Index (Z)			0.041 (0.084)	0.016 (0.035)
Female (NH) × Own DEI Index (Z)			−0.019 (0.079)	0.013 (0.027)
Hispanic × Own DEI Index (Z)			−0.022 (0.072)	−0.030 (0.023)
Male (NH) × Dir Idx (Z) × Own DEI Index (Z)			0.068 (0.068)	0.017 (0.013)
Female (NH) × Dir Idx (Z) × Own DEI Index (Z)			0.096 (0.056)	−0.024 (0.026)
Hispanic × Dir Idx (Z) × Own DEI Index (Z)			−0.166*** (0.053)	0.007 (0.018)
Display Position FE	✓	✓	✓	✓
Resume ID FE	✓	✓	✓	✓
Resume - Coding Test Version FE	×	×	×	×
Respondent ID FE	×	×	×	×
Respondent Age (Cont) and Demographic FEs	✓	✓	✓	✓
Observations	2034	2124	2034	2124
R^2	0.432	0.737	0.436	0.738
Adj. R^2	0.415	0.730	0.417	0.729
R^2 Within	0.001	0.001	0.009	0.002
Adj. R^2 Within	−0.003	−0.003	0.000	−0.005

Dir Idx (Z) = signed directional misalignment index. n = 113 HR, 118 Eng. Coefs flipped post-fit so positive = effect of blinding raises the score. Two-way cluster on responseid and resume index.

Dir Idx (signed) × Own DEI Index — URM-Axis Trichotomy



Top row: HR (high Dir = compensation conditions). Bottom row: Eng (high Dir = rebellion conditions). Bands = 95% CI.

6 Treatment Effect on Productivity of Hired (The Efficiency Myth)

6.1 Does Blinding Lead to More Accurate Coding Skill Evaluations?

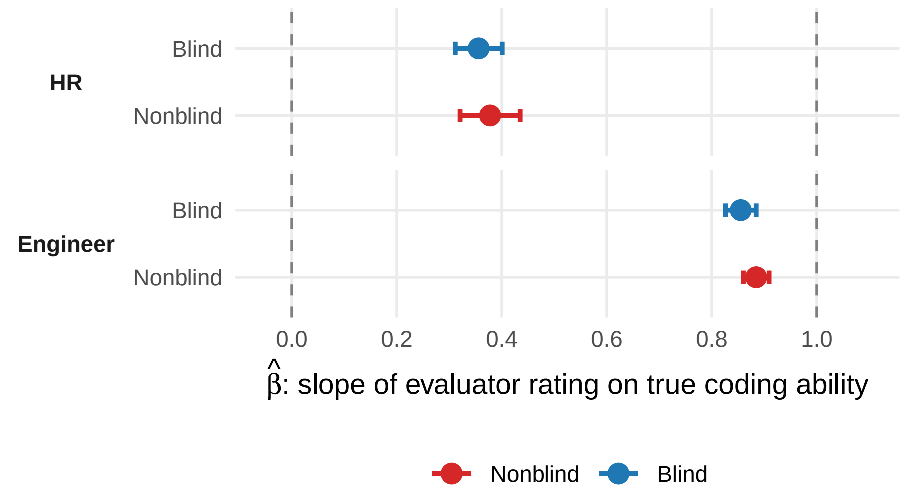
Takeaway: Removing demographic information (or noise) does *not* meaningfully improve evaluators' ability to identify top-ability candidates — signal extraction remains flat.

Specification:

$$\text{Hiring Evaluation Score (Z)} \sim \text{True Coding Ability (Z)} \times \mathbb{1}[\text{Blind arm}] \mid \text{Respondent FE} + \text{Display Position FE}.$$

- $H_0 : \beta = 0$ — the evaluator extracts *no signal* from true coding ability.
- $H_0 : \beta = 1$ — the evaluator is *perfectly calibrated*: a 1 SD rise in true ability produces exactly a 1 SD rise in the rating, with no shrinkage.

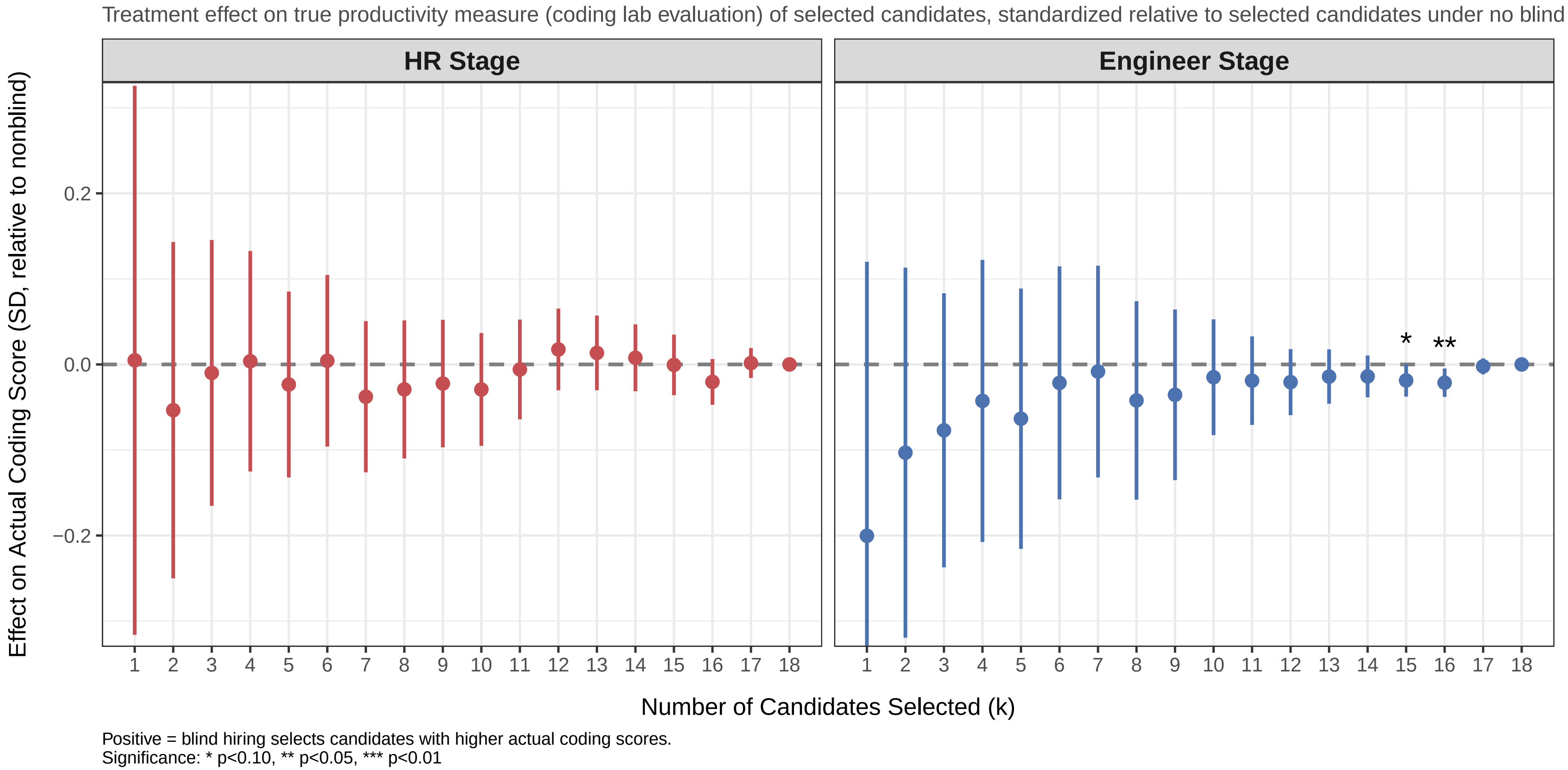
	HR	Engineer
$\hat{\beta}_{\text{nonblind}}$ (slope, nonblind arm)	0.378 (0.029)	0.885 (0.013)
$\hat{\beta}_{\text{blind}}$ (slope, blind arm)	0.356 (0.023)	0.855 (0.015)
Observations	2,466	2,772
R^2	0.142	0.758
Adj. R^2	0.085	0.742
Resume (Content) FE	×	×
Display Position FE	✓	✓
Respondent FE	✓	✓
Respondent Characteristics	×	×



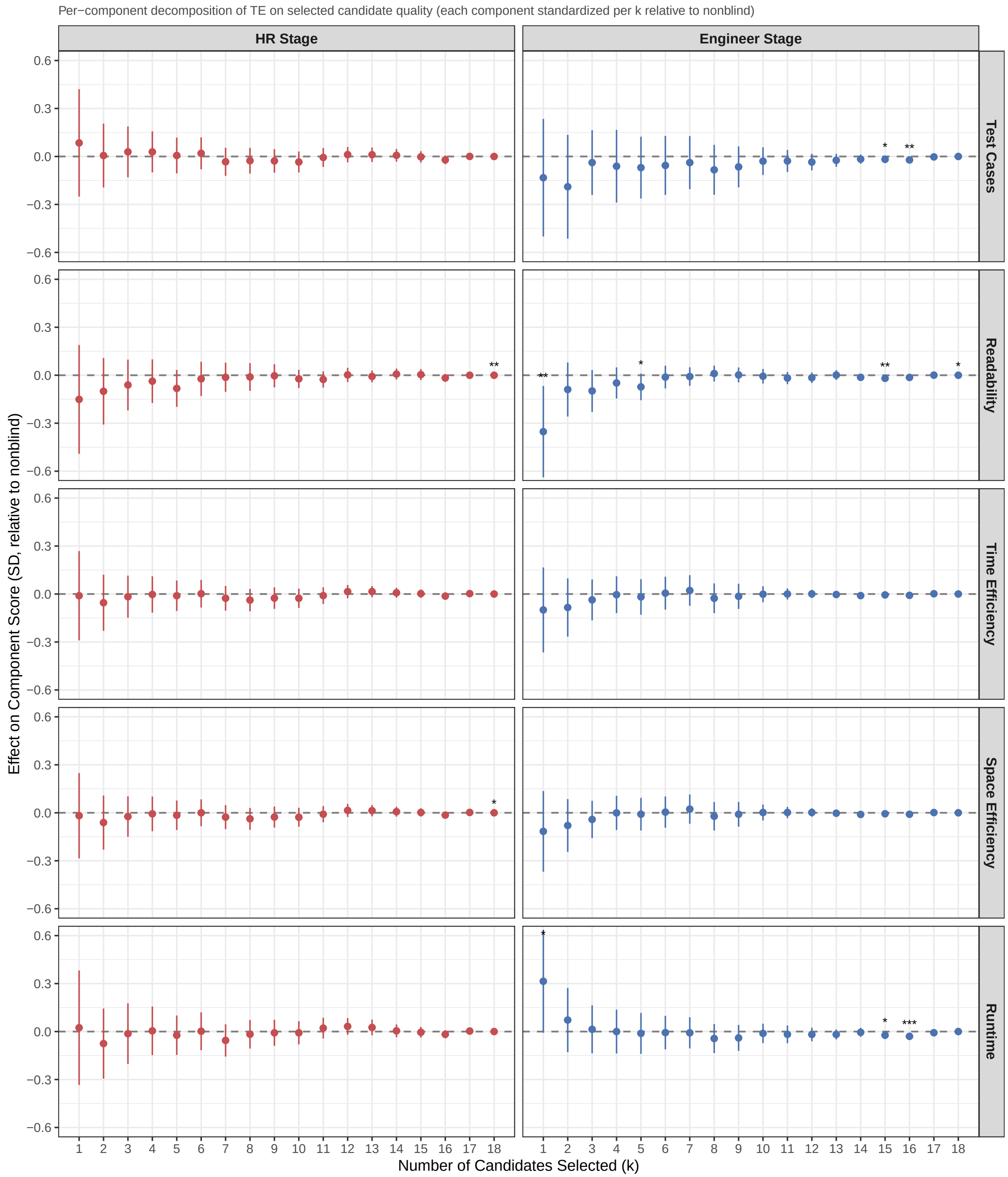
Cluster-robust standard errors (by respondent) shown in parentheses next to each $\hat{\beta}$.
Stars on p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.2 Does Blinding Lead to Selection of More Productive Coders?

Takeaway: Blind hiring has a *null* effect on the actual coding quality of the hired cohort.

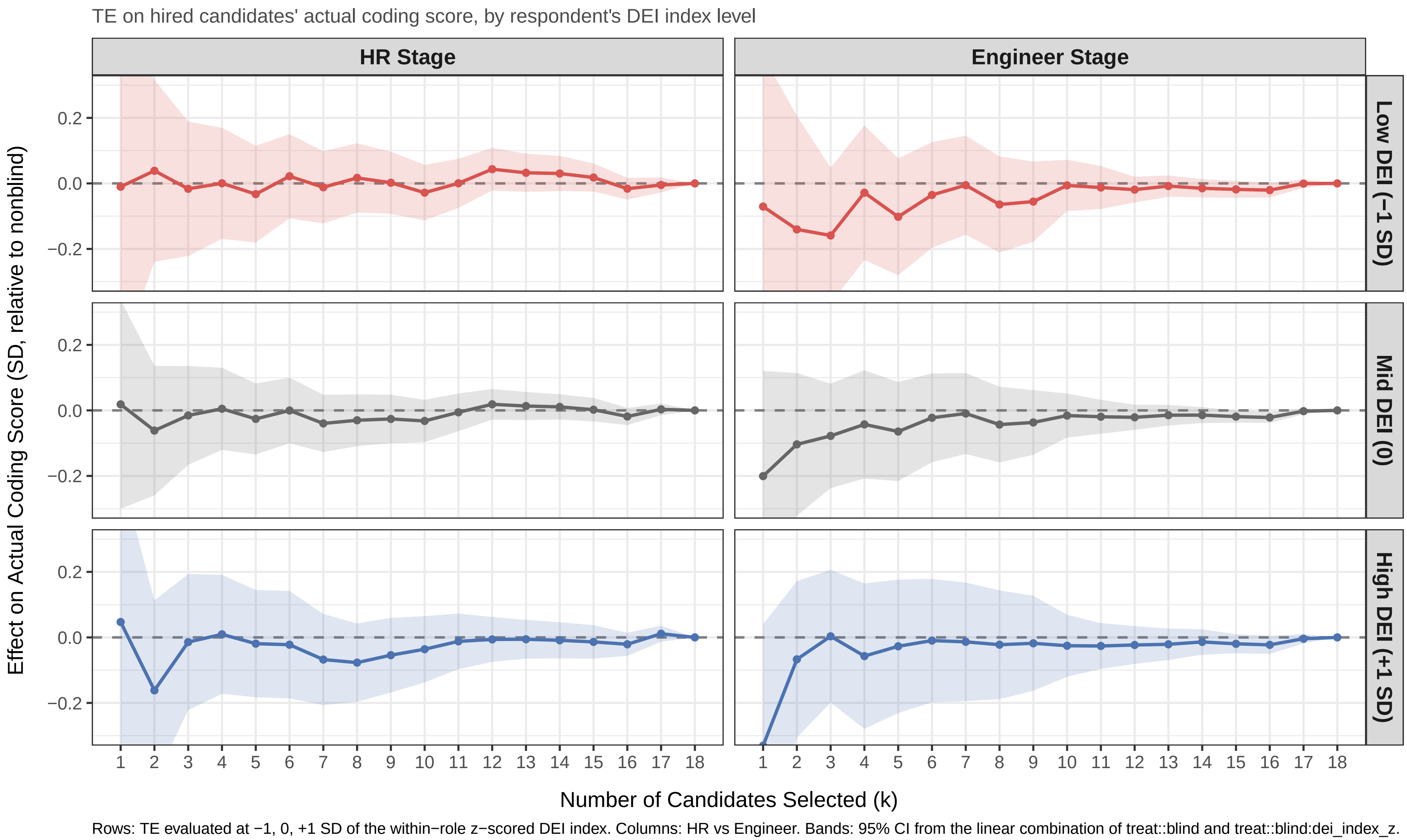


6.2.1 Decomposition of Treatment Effect on Productivity

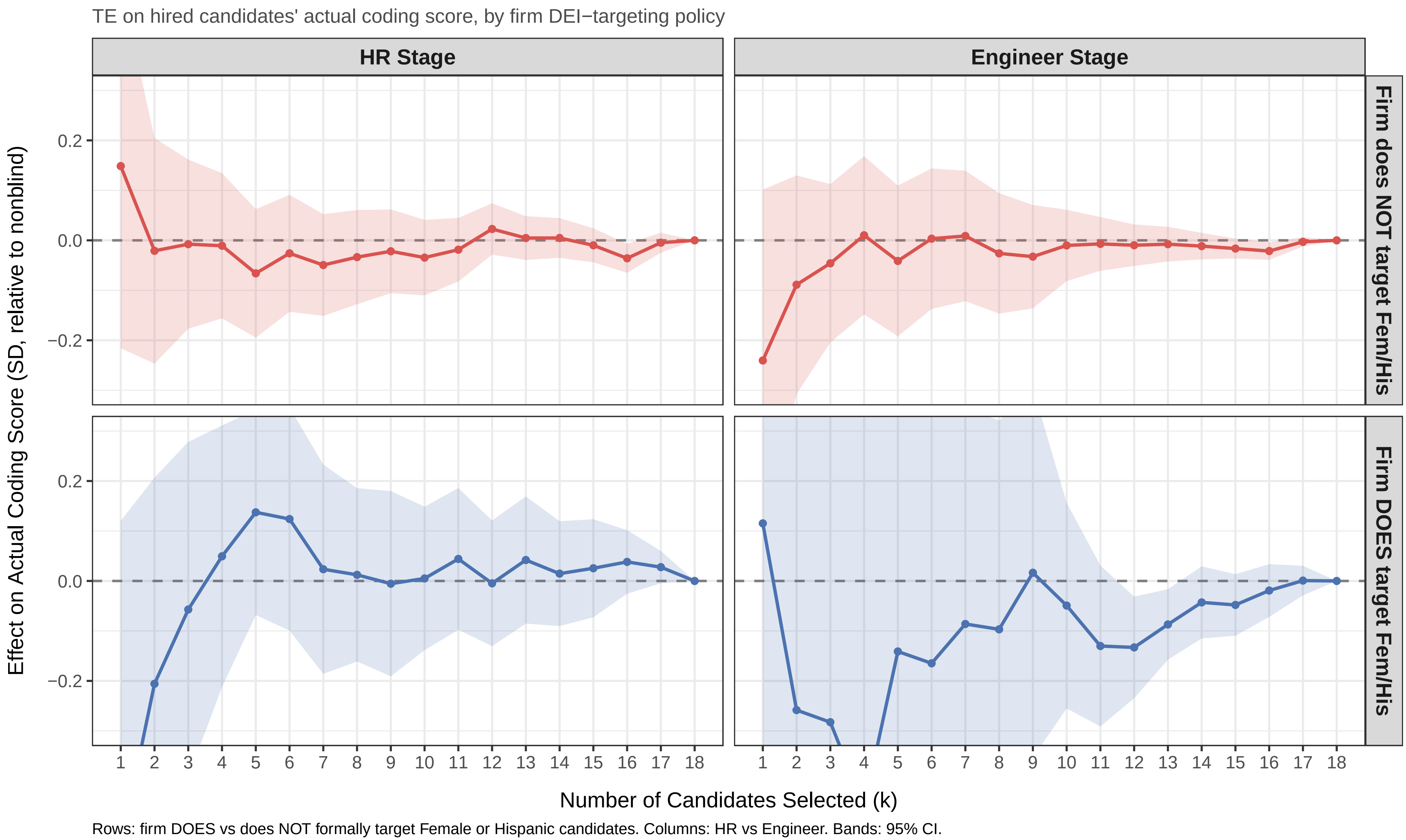


Visibility: engineers see test_case directly; HR sees none. Readability / time / space are unobserved by both sides (latent lab measurements).
Significance: * p<0.10, ** p<0.05, *** p<0.01.

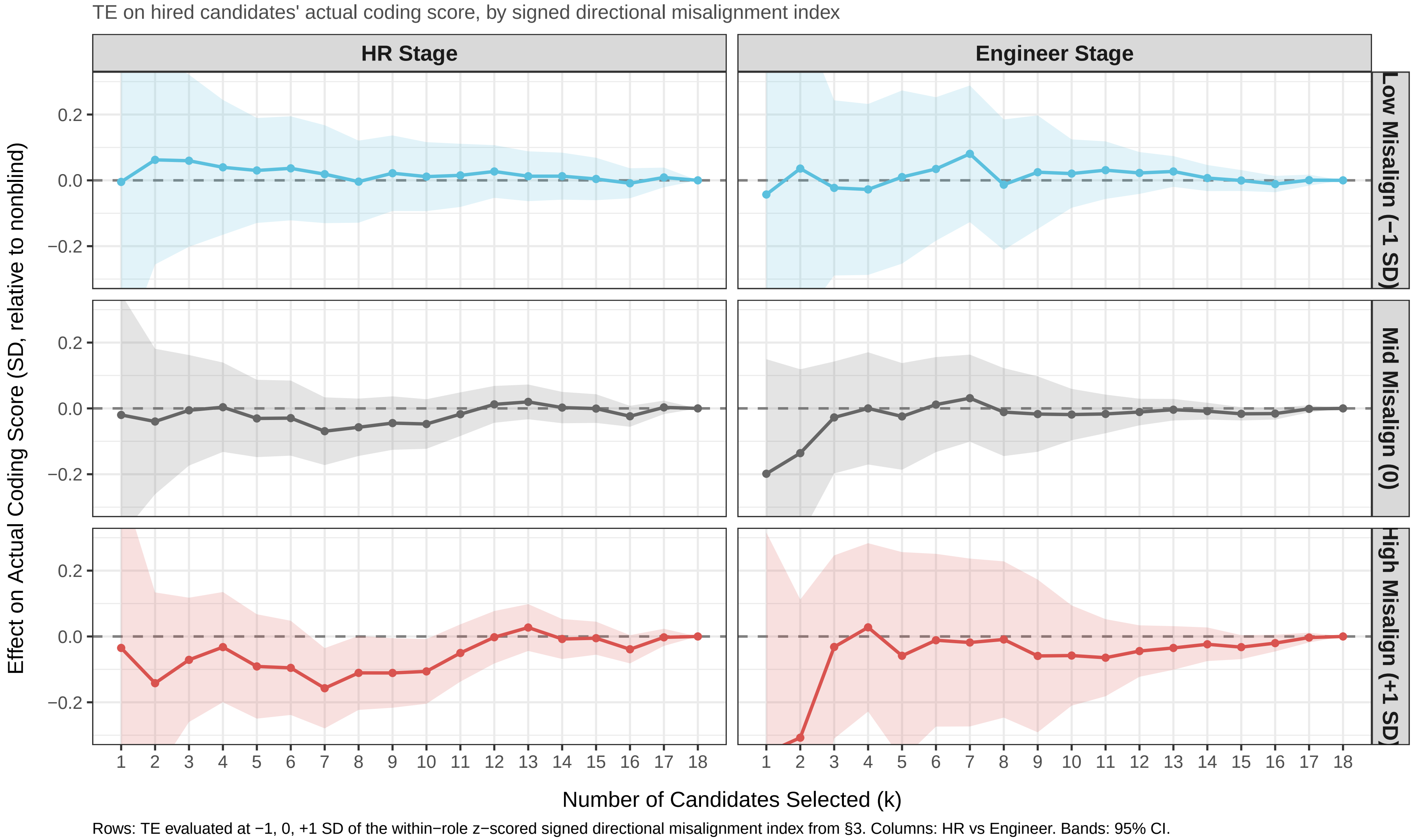
6.3 Does Blinding's Productivity Effect Differ by Evaluator's DEI Orientation?



6.4 Does Blinding's Productivity Effect Differ by Firm DEI-Targeting Policy?



6.5 Does Blinding's Productivity Effect Differ by Inter-Departmental DEI Misalignment?



6.6 Does Blinding's Productivity Effect Vary Jointly by DEI Orientation and Inter-Departmental Misalignment?